

Sentinel — Product Requirements Document

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26 August 2026

Product Requirements Document

Product: Sentinel

Full name: Sentinel Face Attendance Platform

Document type: Product Requirements Document (PRD)

Version: 1.0

Status: Current product (implemented)

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Classification: Internal — product & engineering

1. Purpose of this document

This PRD describes **Sentinel** as a workplace product: what it is for, who uses it, what it must do, and how success is judged.

Sentinel started as a face-recognition lab (FaceNet / ArcFace, LFW evaluation). It is now a **face-attendance platform**: employees check in with a live face, HR sees who is in, and the entrance can run as a kiosk. The recognition research stack remains, but the product promise is **reliable workplace attendance**, not a demo of embeddings.

This document is the source of truth for:

- product scope that exists today
 - roles and permissions
 - user journeys
 - functional and non-functional requirements
 - what is explicitly out of scope
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2. Product summary

One-line	Employees mark attendance with a face. HR sees who is in. The door can run as a kiosk.
Category	Workplace attendance / access control
Primary platform	Web app (React) + FastAPI backend
Local demo	<code>http://localhost:5173</code> (Vite) → API <code>http://127.0.0.1:8000</code>
Default local admin	admin / admin1234 (SQLite)

Tagline (product UI): *Show up. Check in. Go to work.*

3. Problem

Most workplace attendance is still:

- a shared PIN or RFID card (easy to share, easy to lose)
- a paper register or spreadsheet (slow, dishonest, hard to audit)
- a mobile “punch” that can be done from home

Face check-in is stronger **if** it is actually a live person at the workplace. Printed photos, phone screens, and remote punches must fail.

HR also needs a live picture of the day: who is in, who is late, who never got a photo enrolled, and a trail they can export.

4. Goals

4.1 Product goals

1. Check-in is a **face moment**, not a form: camera, oval, optional blink, clear success or retry.
2. HR can run the day without opening a lab: Today dashboard, people directory, reports.
3. The entrance can run **unattended** as a fullscreen kiosk, plus a TV “who is in” screen.
4. Phone check-in can be **locked to the office location** that HR or Admin sets.
5. Recognition quality stays measurable (ROC, TAR@FAR, liveness tester) for operators and admins.

4.2 Non-goals (do not pretend we do these yet)

- Payroll, tax, or salary calculation
- Full leave / PTO workflow and manager approvals
- Mobile native apps (iOS / Android stores)
- Multi-company SaaS tenancy

- Voice or QR as a primary check-in method

5. Users and roles

Role	Who	Home screen	Can do
Employee	Staff member with a linked person record	Check in	Sign in, check in / out, month calendar + CSV, change password
HR	People operations	Today	People, enroll faces, reports, office hours, office location lock, holidays (view), kiosk, activity, TV board
Operator	Reception / security / lab	Recognition lab	Kiosk, activity, TV board, verify / identify / liveness, accuracy charts. Cannot manage people or attendance reports
Admin	IT / founder	Today	Everything HR can, plus staff accounts (create HR / operator / admin, reset passwords), holidays edit, office settings

A person in the directory is not the same as a login. HR must **give a login** and **add a face photo** before that person can check in.

6. User journeys

6.1 New employee

1. Employee creates an account (if public register is on) **or** HR creates the person and login.
2. HR opens **Add faces**, captures a front-facing photo.
3. Employee opens **Check in**. The page waits until a photo exists.
4. Employee looks at the camera, blinks if liveness is required, and is marked present.

6.2 Morning at the door

1. Staff opens **Entrance kiosk** (fullscreen, no sidebar).
2. Visitor or employee faces the oval and taps Check in (blink if required).
3. On match: name, time, late badge; screen resets after a few seconds.
4. If the camera fails: staff uses **employee ID** as a supervised PIN.
5. Optional: a TV on **Entrance screen** (/board) shows who is in now vs still expected.

6.3 HR's day

1. **Today** auto-refreshes: present, late, not in, failed checks, waiting for photo.
2. Click a name → person timeline (punches and events).
3. **Reports** with Today / This week / This month + CSV.
4. Set **office hours**, **weekends**, **holidays**, and **office location lock**.

6.4 Location-locked check-in

1. HR or Admin stands at the workplace.
2. **Today** → **Office location lock** → **Use this device as the office**, pick radius (e.g. 150 m), save.
3. Employee check-in requests GPS. Outside the circle, or location denied → **not present**.
4. Physical **kiosk** at the door does **not** require GPS (the device is already on site).

7. Functional requirements

Priority: **P0** must work for a demo / first workplace; **P1** expected in this build; **P2** later.

7.1 Authentication and accounts — P0

ID	Requirement
AUTH-1	JWT login; session on the SPA.
AUTH-2	Roles: admin, hr, operator, employee.
AUTH-3	Optional public self-register (can be disabled). After register, employee sees a 3-step “what happens next”.
AUTH-4	Admin can create staff users and reset passwords (Staff).
AUTH-5	HR can set or reset an employee username/password from People .
AUTH-6	Employee can change their own password.

7.2 Directory and enrollment — P0

ID	Requirement
PEO-1	Add, search, edit, CSV import, soft-deactivate people.
PEO-2	Fields: name, employee ID (unique), email, notes, department, office, shift start/end.
PEO-3	Enroll one or more face photos; quality gates reject no-face / multi-face / poor frames.
PEO-4	Person timeline: attendance + audit events.
PEO-5	Deactivated people cannot check in; history is kept.

7.3 Employee attendance — P0

ID	Requirement
ATT-1	1:1 verify against the enrolled embedding; threshold from evaluation (TAR@FAR 1e-3 by default).
ATT-2	Optional blink liveness before check-in when <code>require_liveness</code> is on.
ATT-3	One present mark per local calendar day; repeat check-in returns already marked.
ATT-4	Late if after work start + grace (person shift overrides office hours).
ATT-5	Check-out with optional second face; early leave if before work end / shift end.
ATT-6	If office geofence is set: employee check-in must send GPS and be inside the radius.
ATT-7	Personal month view + CSV export.

7.4 Kiosk and TV board — P0 / P1

ID	Requirement
KIO-1	Staff-only fullscreen kiosk; 1:N identify against the gallery.
KIO-2	Oval + scan overlay; success/fail moment; auto-reset ~5s.
KIO-3	Kiosk liveness when required; PIN = employee ID if camera fails.
KIO-4	Kiosk is not GPS-gated.
BRD-1	/board entrance screen: in now, not in yet, live clock, % of expected. TV-friendly, no sidebar.

7.5 HR operations — P0 / P1

ID	Requirement
OPS-1	Today dashboard polls ~12s: present, late, absent, failed, pending faces, still out.
OPS-2	Office hours, timezone, grace, weekend days. HR and Admin may edit.
OPS-3	Office location lock (lat/lng/radius). HR and Admin may edit.
OPS-4	Holidays so expected attendance skips those days.
OPS-5	Reports with date presets and CSV. Activity log with CSV.
OPS-6	Optional email: enrolled face, late, 3 failed checks, absent after start+grace.

7.6 Recognition lab — P1

ID	Requirement
LAB-1	Accuracy charts: ROC, TAR@FAR, score histograms (from reports/evaluation.json).
LAB-2	Verify two photos; identify 1:N.
LAB-3	Blink liveness tester with Live / Not live + blink meter (not raw JSON).

7.7 Platform — P0

ID	Requirement
PLT-1	SQLite locally; Postgres + Alembic in Docker / production.
PLT-2	Rate limits on login and check-in.
PLT-3	Health / ready endpoints. CI: pytest, frontend tests, build.
PLT-4	Dark-first greyscale UI; status badges keep pass/fail colour.

8. Face pipeline (how matching works)

This is the differentiator. Product copy should stay human; this section is for engineering and reviewers.

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Camera frame
→ detect face (SCRFD)
→ quality gates (one face, size, blur, brightness)
→ 5-point align
→ L2-normalised embedding (FaceNet or ArcFace, 512-D)
→ cosine vs enrolled gallery (FAISS inner product or NumPy)
→ match if score ≥ calibrated threshold
```

Encoder	Role
FaceNet (Inception-ResNet-v1, VGGFace2)	Default published weights
ArcFace (w600k_r50 ONNX)	Published reference
Fine-tuned FaceNet + ArcFace head	Optional; evaluated on LFW

Operating point: TAR at FAR = **0.001** on LFW View 2 (6,000 pairs). The live API uses that threshold unless overridden.

Liveness: short blink challenge (eye aspect ratio) plus a texture cue so a printed photo or screen is less likely to pass.

Gallery: embeddings stored per person; identify is 1:N search.

9. Information architecture (screens)

Route	Audience	Purpose
/login	All	Sign in / register
/	HR, Admin	Today — who is in
/board	Staff	TV entrance screen
/people, /people/:id	HR, Admin	Directory and timeline
/enroll	HR, Admin	Face photos
/reports	HR, Admin	Attendance export
/kiosk	Staff	Entrance check-in
/events	Staff	Audit log
/staff	Admin	Create HR / operator / admin
/attendance, /my-attendance, /profile	Employee	Check in, days, password
/lab, /verify, /identify, /liveness	Admin, Operator	Recognition lab

10. Non-functional requirements

Area	Requirement
Latency	Check-in should feel interactive on CPU with a single face (typically a few hundred ms plus network).
Accuracy	Publish LFW accuracy, EER, TAR@FAR; do not silently swap encoders.
Privacy	Store embeddings and audit events, not a video archive of every frame. Limit who can see people and reports.
Security	JWT; production must set a real FRS_SECRET_KEY; disable public register in production; geofence is assistive , not a substitute for a guarded door.
Reliability	/ ready reports database, weights, gallery.
Deploy	Docker Compose + Postgres for a real install; SQLite for local demo.
A11y / UI	Large kiosk type; camera-denied copy; no raw play() errors on the kiosk.
Timezone	Late / absent / “today” use office timezone (default Asia/Kolkata).

11. Success metrics

Metric	Why it matters
Time to first successful check-in (new hire)	Enrollment + login friction
Present vs expected by 10 minutes after start + grace	Did the office actually fill?
False reject rate at the kiosk (qualitative + failed_today)	Light, pose, enrollment quality
False accept incidents	Shared phone, photo attack, PIN misuse
% of employee check-ins blocked by geofence	Policy is working vs GPS too strict
TAR@FAR 1e-3 on LFW	Encoder still honest

12. Risks and mitigations

Risk	Mitigation
Printed photo or replay	Blink + texture liveness; kiosk lighting
Buddy punching from home	Office location lock on phone check-in; kiosk at the door
GPS spoof / coarse GPS	Reasonable radius; kiosk still available; do not claim military-grade geo
Poor enrollment photo	Quality gates; "add another photo"
Threshold too strict / too loose	Lab charts; FAR 1e-3 default
Camera play() races in the browser	Webcam hook ignores abort; single in-flight start
Privacy / labour law	Consent, retention policy, role limits — legal review is on the customer

13. Out of scope / later

These are known and **not** required for this PRD version:

- Leave requests, WFH, manager exception approvals
- Multi-office timezones as first-class tenants (people have an office field; hours are still global)
- Slack alerts (email exists when SMTP is configured)
- PWA install / home-screen check-in as a dedicated track
- Stronger anti-spoof than blink (challenge-response, depth)

- Payroll export beyond attendance CSV

14. Technical snapshot (for implementers)

Layer	Choice
Frontend	React, Vite, Tailwind, Recharts
Backend	FastAPI, SQLAlchemy
Auth	JWT Bearer
DB	SQLite (local), PostgreSQL + Alembic (prod)
Index	FAISS IndexFlatIP
Detect	SCRFD
API prefix	/api/v1

Core resources: /auth/*, /persons/*, /attendance/* (check-in, check-out, kiosk, export), /ops/summary, /settings, /holidays, /users, /events, /liveness/*, /verify, /identify, /metrics.

15. Acceptance criteria (release checklist)

A build matches this PRD when:

1. An employee with a photo can check in; without a photo they cannot.
 2. HR sees Today update without a manual refresh.
 3. Kiosk identifies from the gallery and resets after success/fail.
 4. With location lock **on**, a check-in with no GPS or a far coordinate is rejected; a coordinate inside the radius can succeed.
 5. With location lock **on**, kiosk check-in still succeeds without GPS.
 6. Admin can create an HR user from **Staff**.
 7. Reports CSV and activity CSV download.
 8. Lab charts load when reports/evaluation.json exists.
 9. Backend tests and frontend tests pass.
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16. Glossary

Term	Meaning
Person	Workplace identity (name, employee ID, face embeddings).
User	Login account with a role; employees are linked to one person.
Gallery	Searchable set of enrolled embeddings.
Verify (1:1)	Is this face the claimed person?
Identify (1:N)	Who is this face among enrolled people?
TAR / FAR	True-accept rate / false-accept rate.
Liveness	Evidence the camera sees a live person (blink), not a photo.
Geofence	Circle around the office; phone check-in must fall inside it.
Kiosk	Staff-supervised entrance device, fullscreen.
Entrance screen	TV board at /board showing who is in.

17. Document control

Version	Date	Notes
1.0	26 Aug 2026	First full PRD for Sentinel as a face-attendance product, matching the implemented app.

Owner: Product / project lead

Reviewers: Engineering, HR operations (customer)

Related: README.md (setup and evaluation), reports/TAR_FAR_REPORT.md (encoder numbers)